QUANTUM D-MODULES OF PROJECTIVE BUNDLES AND BLOWUPS

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Abstract. In this lecture series, we review the basics of quantum D-modules and Givental's formalism. We then explain the projective bundle formula and the blowup formula, which are proved using Fourier transforms associated with rank-one torus actions. If time permits, we will also discuss a nonabelian version of the Fourier transform.

Contents

Motivation 1
1. Quantum D-modules 1
2. Givental formalism 3

Motivation

One motivation for us comes from mirror symmetry. Let X be a Fano variety (or more generally, log Calabi-Yau). Then we expect there to be a mirror $\check{X}$, which is an LG model. Via some version of Saito theory (which may not actually be well-defined), we obtain a Gauss-Manin conjecture. This will correspond to the quantum D-module of X, which is defined using Gromov-Witten theory.

Another motivation comes from the study of other areas of algebraic geometry[1]. Katzarkov-Kontsevich-Pantev-Yu define Hodge atoms which establish an irrationality criterion for varieties and was used to prove the irrationality of the cubic fourfold. Recently, an unreleased internal model at OpenAI proved the generalized Mukai conjecture using Gromov-Witten theory.

1. Quantum D-modules

Let X be a smooth projective variety and $(\phi_i)_i$ be a homogeneous basis with corresponding coordinates $(\tau^i)_i$. Then we define the (*big*) *quantum cohomology*

$$\mathrm{QH}^*(X) \coloneqq (H^*(X)[\![Q,\tau]\!],\star),$$

where $\mathbb{C}[Q] \coloneqq \mathbb{C}[\mathrm{NE}_{\mathbb{N}}(X)]$ keeps track of effective curve classes, and here we are taking a graded completion of $\mathbb{C}[Q,\tau]$ where

$$\deg Q^\beta = 2\int_\beta c_1(X) \qquad \text{and} \qquad \deg \tau^i = 2 - \deg \phi_i.$$

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[1]Which we have historically been quite isolated from for various reasons

Example 1.1. Let $X = \mathbb{P}^n$ and $\tau = 0$. Then the curve classes are just non-negative integers, so Q has degree $2(n+1)$. In this case, $\mathbb{C}[\![Q]\!] = \mathbb{C}[Q]$.

Example 1.2. Let X have $\mathbb{C}[Q] = \mathbb{C}[Q_1, Q_2]$ where $\deg Q_1 = 2$ and $\deg Q_2 = -2$. In this case, we actually have

$$\mathbb{C}[\![Q_1, Q_2]\!] = \bigoplus_{n\geq 0} Q_1^n \mathbb{C}[\![Q_1Q_2]\!] \oplus \bigoplus_{n<0} Q_2^n \mathbb{C}[\![Q_1Q_2]\!],$$

which lives in all even degrees.

Now let $\alpha, \beta, \gamma \in H^*(X)$. The quantum product is defined by the formula

$$(\alpha \star \beta, \gamma)^X := \sum_{d,n} \frac{Q^d}{n!} \langle \alpha, \beta, \gamma, \tau, \dots, n \rangle_{0,n+3,d}^X,$$

where $(-,-)^X$ is the Poincaré pairing and $\tau = \sum_i \tau^i \phi_i$.

Remark 1.3. We can define the quantum cohomology as a formal Frobenius manifold, but this is equivalent to the approach taken by the speaker, so we will skip it.

Definition 1.4. We will now define the quantum D-module

$$\mathrm{QDM}(X) := (H^*(X)[z][\![Q, \tau]\!], \nabla^X, P_X).$$

The connection ∇^X is known as the *quantum connection* and is defined by

$$\begin{aligned} \nabla^X_{z\xi Q\partial_Q} &:= z\xi Q\partial_Q + \xi \star -, \\ \nabla^X_{z\partial_{\tau^i}} &:= z\partial_{\tau^i} + \phi_i \star - \\ \nabla^X_{z^2\partial_z} &:= z^2\partial_z - (E_X \star -) + z \cdot \mu_X. \end{aligned}$$

Here, the first definition makes sense only for $\xi \in H^2(X)$ and the *Euler vector field* is defined by

$$E_X := c_1(X) + \sum_i \left(1 - \frac{\deg \phi_i}{2}\right) \tau^i \phi_i.$$

Additionally, the *grading operator* is defined by $\mu_X(\phi_i) = \frac{\deg \phi_i - \dim X}{2} \phi_i$. Finally, the pairing

$$P_X := \mathrm{QDM}(X)^{\otimes 2} \to \mathbb{C}[z][\![Q, \tau]\!]$$

is given by the formula

$$P_X(f(z), g(z)) := \int_X f(-z) \cup g(z).$$

Example 1.5. Let $X = \mathbb{P}^1$. Then we have

$$\mathrm{QH}^*(\mathbb{P}^1) \cong \frac{\mathbb{C}[p, Q, \tau^0][\![\tau^1]\!]}{p^2 - Qe^{\tau^1}}$$

and

$$\mathrm{QDM}(\mathbb{P}^1) = H^*(\mathbb{P}^1)[z, Q, \tau^0][\![\tau^1]\!]$$

with the relevant connections.

Remarks 1.6.

(1) The object $\mathrm{QDM}(X)$ is not a D-module in the usual sense of D-modules.[2]

[2] Maybe it is an R-module?

(2) The objects QH and QDM contain the same information, namely that one can be recovered from the other.
(3) The ring $\mathrm{QH}^*(X)$ is a quantization of the usual cohomology ring $H^*(X)$ as a ring and $\mathrm{QDM}(X)$ is a quantization of $H^*(X)$ as a graded $H^*(X)$-module.

We will now produce the fundamental solution $M_X(\tau)$ to the quantum connection. This is given by the formula

$$(M_X(\tau)\alpha,\beta)^X = (\alpha,\beta)^X + \sum_{(n,d\neq(0,0))} \frac{Q^d}{n!}\left\langle \frac{\beta}{z-\psi}, \tau, \dots, \tau, \alpha \right\rangle^X_{0,n+2,d}$$

for all $\alpha,\beta \in H^*(X)$.

Proposition 1.7. *We have*

$$\begin{vmatrix} z\xi Q\partial_Q + \xi\cup \\ z\partial_{\tau^i} \\ z^2\partial_z - c_1(X) + z\cdot\mu_X \end{vmatrix} \circ M_X(\tau) = M_X(\tau) \circ \begin{vmatrix} z\xi Q\partial_Q + \xi\star \\ z\partial_{\tau^i} + \phi_i\star \\ z^2\partial_z - E_X\star + z\cdot\mu_X \end{vmatrix}.$$

Moreover, $M_X(\tau)1 = J_X(\tau)$ *and* $M_X(\tau)\phi_i = z\partial_{\tau^i}J_X(\tau)$. *We also have the asymptotics*

$$M_X(\tau) = e^{\tau/z}\mathrm{id} + O(Q)$$

and the property

$$(M_X(\tau)\alpha, M_X(\tau)|_{z\mapsto -z}\beta)^X = (\alpha,\beta)^X.$$

2. Givental formalism

Define

$$(z) = \sum_{k\geq 0}\sum_i t^i_k\phi_i z^k$$

and the genus zero descendent GW potential

$$\mathscr{F}_X(z1 + (z)) \coloneqq \sum_{n,d}\frac{Q^d}{n!}\langle (-\psi),\dots,(-\psi)\rangle^X_{0,n,d}.$$

Now define the symplectic vector space

$$\mathscr{H}_X \coloneqq H^*(X)[z^{\pm}]\llbracket Q\rrbracket.$$

This is actually functorial, but we will skip the discussion of this. Then define the form

$$\Omega_X\colon \mathscr{H}_X\times\mathscr{H}_X \to \mathbb{C}\llbracket Q\rrbracket$$

by the formula

$$\Omega_X(f(z),g(z)) \coloneqq [z^{-1}](f(-z),g(z))^X.$$

This makes $\mathscr{H}_X$ an infinite-dimensional symplectic vector space over $\mathbb{C}\llbracket Q\rrbracket$.

There is a decomposition

$$\mathscr{H}_X = \mathscr{H}_+ \oplus \mathscr{H}_-,$$

where

$$\mathscr{H}_+ = H^*(X)[z]\llbracket Q\rrbracket \qquad \text{and} \qquad \mathscr{H}_- = z^{-1}H^*(X)[z^{-1}]\llbracket Q\rrbracket.$$

Finally, we can define $\mathscr{L}_X$ to be the graph of $\mathrm{d}\mathscr{F}_X$ and refer to it as the *Givental cone*.

Theorem 2.1. $\mathscr{L}_X \subset [H]_X$ *is an overruled Lagrangian cone; namely:*

(1) $\mathscr{L}_X \subset (\mathscr{H}_X,\Omega_X)$ *is Lagrangian;*
(2) *It is a cone, namely that if* $I(\tau)\in\mathscr{L}_X$ *and* $F(\tau,z)I(\tau)\in\mathscr{L}_X$ *if* $F(\tau,z) = 1 + O(\tau)$;

(3) *For any* $f \in \mathscr{L}_X$, *we have* $T_f\mathscr{L}_X \cap \mathscr{L}_X = zT_f\mathscr{L}_X$.

Insert figure here.

Example 2.2. When $X = \mathbb{P}^1$, we see that the basis $1, p, z, pz, z^2, \ldots$ is dual to the basis $z^{-1}p, z^{-1}, z^{-2}p, z^{-2}, z^{-3}p, \ldots$. When we multiply by z, we see that $\mathscr{H}_-$ is not preserved.

Remark 2.3. Let F be a formal function on $\mathscr{H}_+$. Then the graph $\mathscr{L}$ of $\mathrm{d}F$ is an overruled Lagrangian cone if and only if F satisfies the string equation, dilaton equation, and topological recursion relations.

There is an informal description of the Lagrangian cone as

$$\mathscr{L}_X = \bigcup_{t \in H^*(X)[\![Q]\!]} zT_t\mathscr{L}_X \coloneqq M_X(t)\mathscr{H}_+.$$

In particular, we see that

$$M_X(\tau)\colon \mathrm{QDM}(X) \to T_\tau\mathscr{L}_X$$

is an isomorphism intertwining ∇ and d. This essentially explains all of genus zero Gromov-Witten theory.

Table 1. Comparison

	QDM(X)	$T_\tau\mathscr{L}_X$
differential operator	hard	easy
description as $\mathbb{C}[z][\![Q, \tau]\!]$-module	easy	hard

For example, if we consider

$$\begin{array}{ccc} \mathrm{QDM}(\mathbb{P}(V)) & \longrightarrow & \mathrm{QDM}(B)^{\oplus N} \\ \downarrow{\scriptstyle M_{\mathbb{P}(V)}} & & \downarrow{\scriptstyle M_B} \\ T\mathscr{L}_{\mathbb{P}(V)} & \longrightarrow & (T\mathscr{L}_B)^{\oplus N}, \end{array}$$

we prove that the bottom arrow is an isomorphism, then go around the diagram to obtain the top arrow.